

UCS503 – Software Engineering Project Report

Sports Resource Management System

A Mobile Application for Sports Resource and Facility
Management

Submitted by:

1024240041 – Soumya Smita Sahoo

1024240044 – Gaganpreet Kaur

1024240038 – Himasri Chadavalavada

Group: 7

BE Third Year, CSE

Submitted to:

Ms. Anushka Chaudhary

Mid-Semester Evaluation

Computer Science and Engineering Department

Thapar Institute of Engineering and Technology, Patiala

September 8, 2026

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Introduction

1.1 Background and Context

Sports facilities and equipment in educational institutions need to be managed efficiently so that students, coaches, sports officers, and maintenance staff can access and manage resources in an organized manner. When such activities are handled manually, it can become difficult to track equipment availability, facility bookings, equipment requests, and maintenance activities.

The proposed Sports Resource Management System is a mobile application designed to provide a centralized platform for managing sports resources and facilities. The application allows users to access different functionalities according to their roles.

The system is developed using Flutter and Dart for the mobile application, with a Node.js backend and PostgreSQL database. The application provides role-based functionality for students/players, coaches, sports officers, administrators, and maintenance staff.

1.2 Problem Statement

Managing a large number of sports facilities and equipment manually can result in difficulties in tracking availability, bookings, requests, and maintenance records.

Students and coaches may need to know whether a facility or equipment is available before making a request or booking. Sports management personnel also require a centralized method to manage equipment and facilities. Maintenance staff need to identify the exact equipment that is under repair and track its maintenance status.

Therefore, the project aims to develop a mobile application that provides an organized and centralized solution for sports resource and facility management.

1.3 Project Objectives

The main objectives of the Sports Resource Management System are:

- To provide a centralized mobile platform for managing sports resources and facilities.
- To allow students and coaches to view the availability of sports facilities.
- To provide facility booking functionality.

- To allow users to request sports equipment.
- To maintain equipment information such as equipment ID, category, quantity, condition, and status.
- To provide a mechanism for tracking equipment maintenance.
- To identify specific equipment under repair using equipment IDs.
- To maintain booking and equipment request history.
- To provide role-based access for different categories of users.
- To provide feedback and notification functionality as part of the overall system design.

1.4 Scope

The scope of the project is the development of a mobile-based Sports Resource Management System for managing sports facilities and equipment.

The system considers the following user roles:

- Student / Player
- Coach
- Sports Officer
- Administrator
- Maintenance Staff

Students and coaches can interact with facilities, bookings, equipment, equipment requests, and related services. Sports officers can manage sports resources and bookings. Maintenance staff are responsible for equipment maintenance and repair-related activities.

The current work is an initial working prototype. Therefore, some functionalities are still under development and complete system evaluation and final results are not included in this report.

Methodology and Design

2.1 System Architecture

The Sports Resource Management System follows a client-server architecture. The mobile application acts as the client, while the Node.js backend handles communication between the application and the PostgreSQL database.

The Flutter application provides the user interface and role-based interaction. The backend processes requests and communicates with the database for storing and retrieving system information.

2.1.1 High-Level Design

The major components of the system are:

- Flutter mobile application
- Node.js backend server
- PostgreSQL database
- Role-based user access
- Sports resource management modules

The user interacts with the Flutter application. Depending on the selected role and login credentials, the user is provided with the corresponding functionalities. The backend processes the application requests and interacts with the PostgreSQL database.

2.1.2 Role-Based Access

The system is designed to support multiple roles.

- **Student/Player:** Can view resources, check availability, book facilities, request equipment, view request status, and access relevant history.
- **Coach:** Can access facility booking and equipment-related functionalities required for sports activities.
- **Sports Officer:** Can manage sports facilities, equipment, bookings, equipment requests, and related records.

- **Administrator:** Provides administrative management of users, roles, permissions, and system-level information.
- **Maintenance Staff:** Handles equipment repair and maintenance status and can identify equipment using equipment IDs.

2.2 Technical Stack

The technologies used for the project are:

- **Frontend Framework:** Flutter
- **Programming Language:** Dart
- **Backend:** Node.js
- **Database:** PostgreSQL
- **Development Environment:** Visual Studio Code
- **Version Control:** GitHub

2.3 Database Design

The PostgreSQL database is designed to store information related to users, equipment, bookings, facilities, equipment requests, maintenance, feedback, and notifications.

The major database entities are:

- User
- Equipment
- Booking
- Facility
- Equipment Request
- Maintenance
- Feedback
- Notification

These entities are connected through relationships representing the different activities performed within the system.

2.3.1 Equipment Identification and Maintenance Tracking

Each equipment item is associated with a unique equipment ID. This ID is used to identify the particular equipment item when it is issued, returned, or sent for maintenance.

When an equipment item requires repair, its equipment ID can be associated with the corresponding maintenance record. This allows maintenance staff to determine which particular equipment is under repair rather than relying only on the equipment name or category.

2.4 UML and System Models

2.4.1 Use Case Diagram

The Use Case Diagram represents the interactions between the different actors and the Sports Resource Management System.

The major actors considered are Student/Player, Coach, Sports Officer, Administrator, and Maintenance Staff.

The diagram represents activities such as login, role selection, viewing equipment availability, booking facilities, requesting equipment, managing equipment, managing facilities, handling maintenance, viewing history, providing feedback, and receiving notifications.

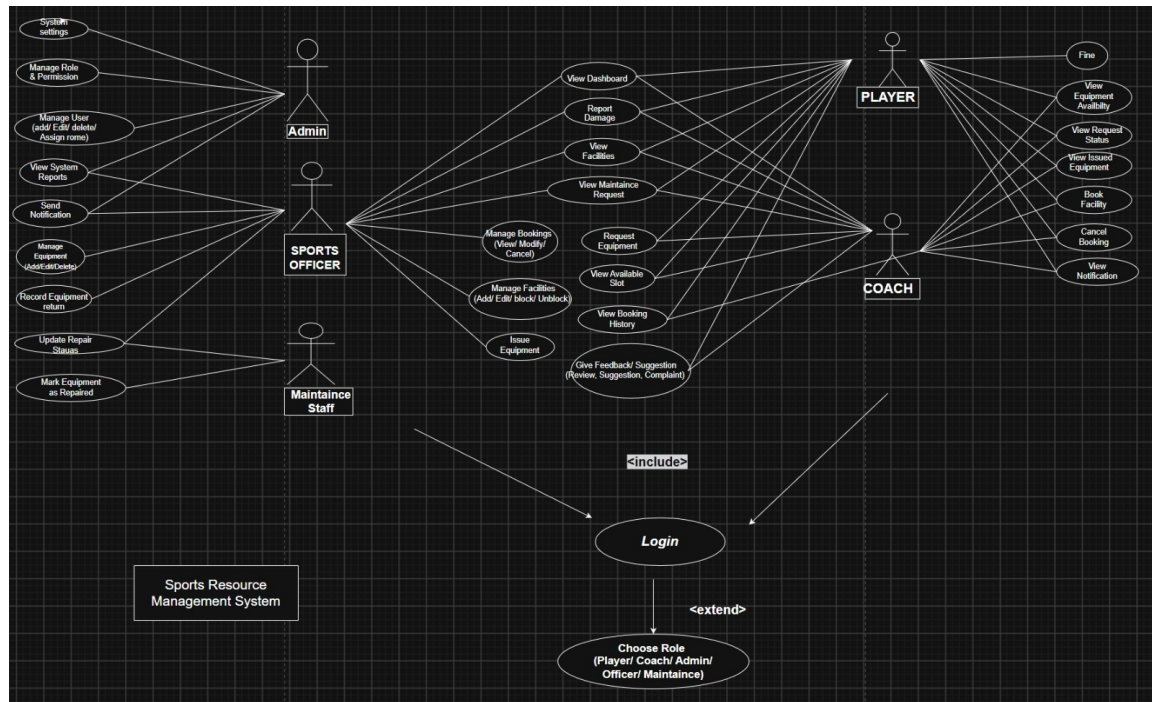


Figure 2.1: Use Case Diagram

2.4.2 Class Diagram

The Class Diagram represents the major entities of the Sports Resource Management System and their relationships.

The main classes represented in the design are User, Equipment, Booking, Facility, Equipment Request, Maintenance, Feedback, and Notification.

The relationships also represent the multiplicity between entities. For example, a user can have multiple bookings and multiple equipment requests. Equipment can have multiple maintenance records over its lifetime.

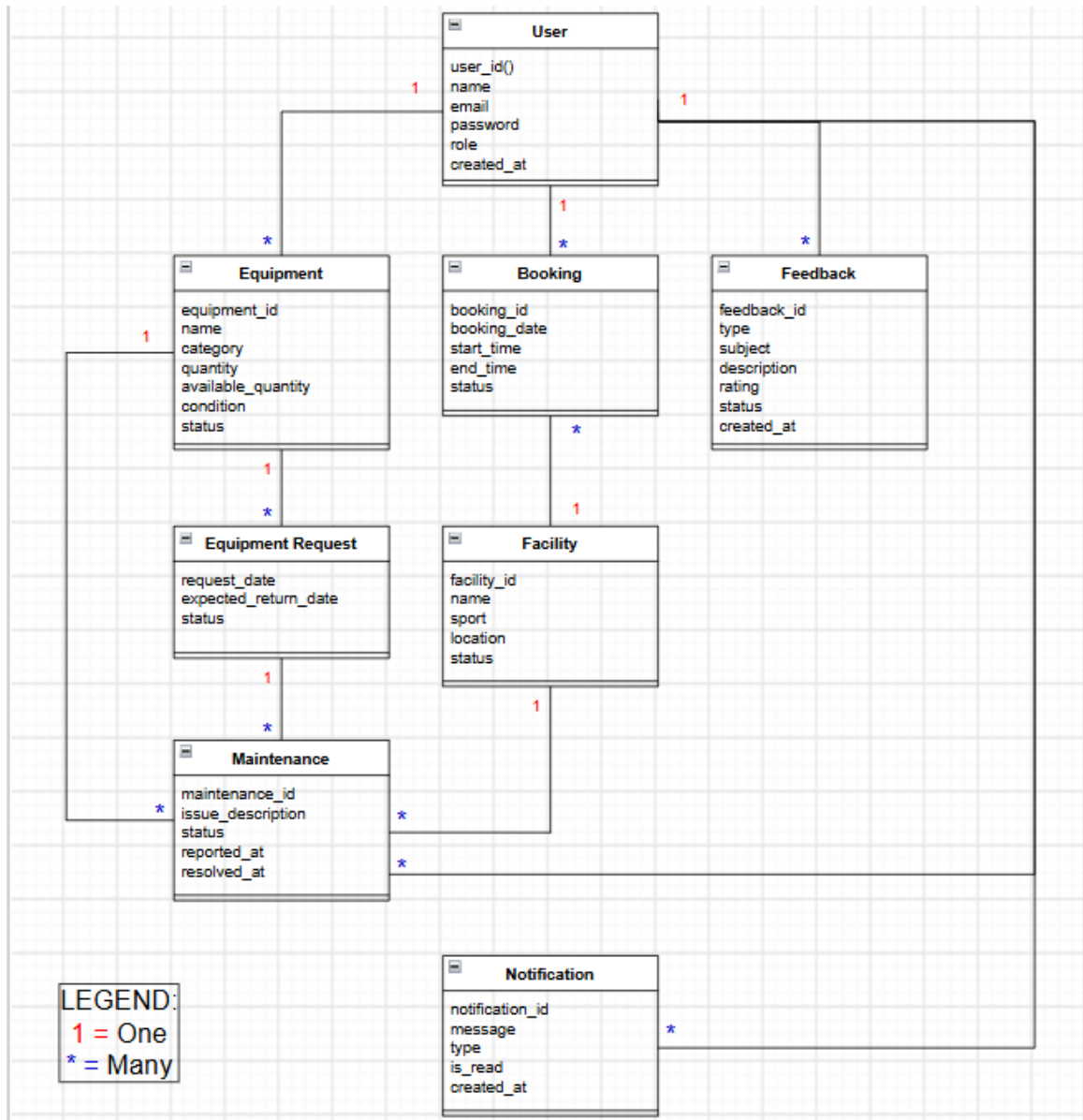


Figure 2.2: Class Diagram

2.4.3 Activity Diagram

The Activity Diagram represents the workflow of major operations in the system.

The facility booking workflow includes viewing facility availability, selecting a date and time, submitting a booking request, checking for conflicts, and confirming the booking when a slot is available.

The system also considers the waiting-list workflow when the requested facility slot is not available.

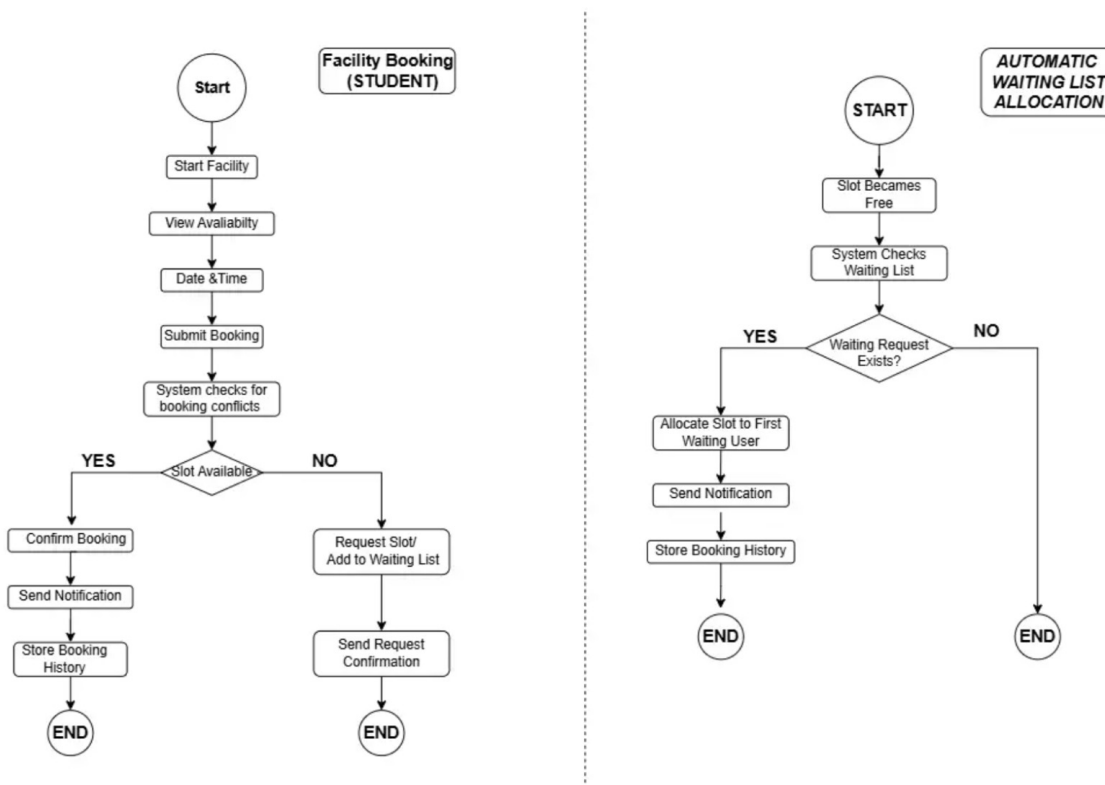


Figure 2.3: Activity Diagram

2.5 Implementation Details

2.5.1 Module 1: User Authentication and Role Selection

The application begins with role selection. The user can select the appropriate role before proceeding to the login interface.

The role-based structure allows different users to access the functionality relevant to their responsibilities.

The authentication and role-selection flow forms the entry point of the application.

2.5.2 Module 2: Equipment Management

The equipment management module maintains information about sports equipment.

The equipment information includes:

- Equipment ID
- Equipment name
- Category
- Quantity
- Available quantity
- Condition
- Status

The equipment ID is particularly important for identifying individual equipment during issue, return, and maintenance operations.

2.5.3 Module 3: Facility Booking

The facility booking module allows users to view available facilities and select suitable dates and time slots.

The booking information includes booking ID, booking date, start time, end time, and booking status.

The system checks whether a requested slot is available before confirming the booking.

2.5.4 Module 4: Equipment Requests

The equipment request module allows users to submit requests for sports equipment.

The request stores information such as request date, expected return date, and request status.

The request status can be used to track the progress of the equipment request.

2.5.5 Module 5: Maintenance

The maintenance module is designed to handle equipment-related repair activities.

A maintenance record contains information such as maintenance ID, issue description, status, reported date, and resolved date.

Equipment IDs are used to associate maintenance activities with the specific equipment item being repaired.

The maintenance staff interface is part of the planned system but is currently under development in the prototype.

2.5.6 Module 6: Feedback and Notifications

The system design includes feedback and notification functionality.

The feedback module is intended to allow users to provide reviews, suggestions, complaints, and ratings.

The notification module is intended to communicate relevant system events to users, such as booking-related updates and request status changes.

These functionalities are currently under development in the prototype.

Current Prototype Status

3.1 Working Features

The current prototype contains the core application structure and several working functionalities.

The implemented portions include:

- Flutter-based mobile application interface
- Role selection and login flow
- Student/Player functionality
- Coach functionality
- Sports Officer functionality
- Equipment-related functionality
- Facility-related functionality
- Facility booking
- Equipment request functionality
- Database connectivity
- PostgreSQL database tables
- Node.js backend

3.2 Features Under Development

The following functionalities are currently not fully working and are under development:

- Administrator functionality
- Maintenance staff interface
- Feedback functionality

- Notification functionality

These features have been included in the system design but require further implementation and testing.

3.3 Current Development Structure

The project is organized into separate application and backend components.

The Flutter mobile application is contained in the `mobile_app` directory. The application source code is primarily in Dart.

The backend is contained in the `backend` directory. It contains the Node.js server, database connection, user creation scripts, and package configuration files.

The PostgreSQL database contains the tables required by the system.

The source code is maintained using GitHub for version control.

3.4 Prototype Limitations

The current work represents an initial working prototype prepared for mid-semester evaluation.

Some modules are still under development and therefore are not considered complete.

The following aspects are intentionally not presented as final results:

- Complete performance evaluation
- Final testing results
- User acceptance results
- Deployment results
- Complete administrator functionality
- Complete maintenance staff functionality
- Complete feedback implementation
- Complete notification implementation

These aspects can be addressed during further development of the project.

Conclusion

4.1 Summary of Work

The Sports Resource Management System is a mobile application designed to provide a centralized platform for managing sports facilities and equipment.

The application is developed using Flutter and Dart, with a Node.js backend and PostgreSQL database.

The system provides different functionalities according to user roles and supports operations such as facility booking, equipment management, and equipment requests.

The project also incorporates equipment identification through equipment IDs to support tracking and maintenance operations.

4.2 Key Findings

The initial prototype demonstrates the feasibility of using a mobile application to organize sports resource management activities.

The role-based structure allows the system to provide different functionalities to different categories of users. The database design also provides a structured way of storing information related to equipment, facilities, bookings, requests, and maintenance.

4.3 Future Work and Recommendations

Future development of the system will focus on completing the currently unfinished modules.

The planned improvements include:

- Complete administrator functionality
- Complete maintenance staff interface
- Implement equipment maintenance tracking
- Implement feedback functionality
- Implement notification functionality
- Improve booking and request tracking

- Perform complete system testing
- Conduct performance evaluation
- Improve the overall user interface and experience

4.4 Final Remarks

The current prototype establishes the basic foundation of the Sports Resource Management System.

The implemented modules demonstrate the main concept of the proposed application, while the remaining modules can be completed during the next stages of development.

Further implementation and testing will be required before the system can be considered a complete production-ready application.